

ELITE IGCSE ACADEMY

Student Past Paper Solutions

May-Nov 2020 P1H Worked Solutions

MayNov 2020 | P1H

31 questions ■ question image followed by worked solution

PREPARED BY

Dr. Eslam Ahmed

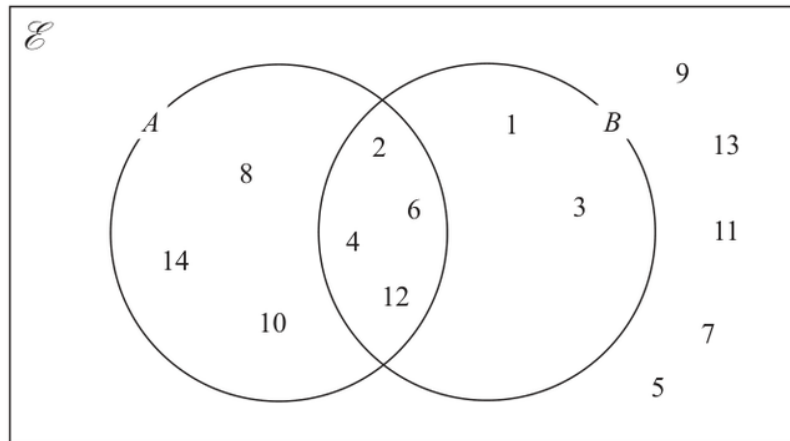
Assistant Lecturer, Cairo University Faculty of Engineering

WhatsApp: +20 112 000 9622 | eliteigcse.com

PAST PAPER QUESTION**Q#: 1****Marks: 4****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

May-Nov 2020 P1H - MayNov 2020 | P1H

- 1 The numbers from 1 to 14 are shown in the Venn diagram.



- (a) List the members of the set $A \cap B$

.....
(1)

- (b) List the members of the set B'

.....
(1)

A number is picked at random from the numbers in the Venn diagram.

- (c) Find the probability that this number is in set A but is **not** in set B.

.....
(2)

.....
(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION**Q#: 1****Marks: 4****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate probability**

From the Venn diagram,

$$A \cap B = \{2, 4, 6, 12\}$$

2. The values not in areThe values not in B are

$$B' = \{5, 7, 8, 9, 10, 11, 13, 14\}$$

3. The values in but notThe values in A but not in B are

$$\{8, 10, 14\}$$

4. Calculate probability

So

$$P(A \cap B') = \frac{3}{14}$$

FINAL ANSWER $\{2, 4, 6, 12\}, \{5, 7, 8, 9, 10, 11, 13, 14\}, \frac{3}{14}.$

PAST PAPER QUESTION**Q#: 2****Marks: 4****TOPIC: Probability Toolkit**

May-Nov 2020 P1H - MayNov 2020 | P1H

- 2** Toy cars are made in a factory.
The toy cars are made for 15 hours each day.
5 toy cars are made every 12 seconds.

For the toy cars made each day, the probability of a toy car being faulty is 0.002

Work out an estimate of the number of faulty toy cars that are made each day.

(Total for Question 2 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 2****Marks: 4****TOPIC: Probability Toolkit**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate probability**

$$15 \text{ hours} = 15 \times 60 \times 60 = 54000 \text{ seconds}$$

2. The machine makes cars every

The machine makes 5 cars every 12 seconds, so it makes

$$\frac{54000}{12} \times 5 = 22500$$

3. cars

cars.

4. Expected faulty cars

Expected faulty cars:

$$22500 \times 0.002 = 45$$

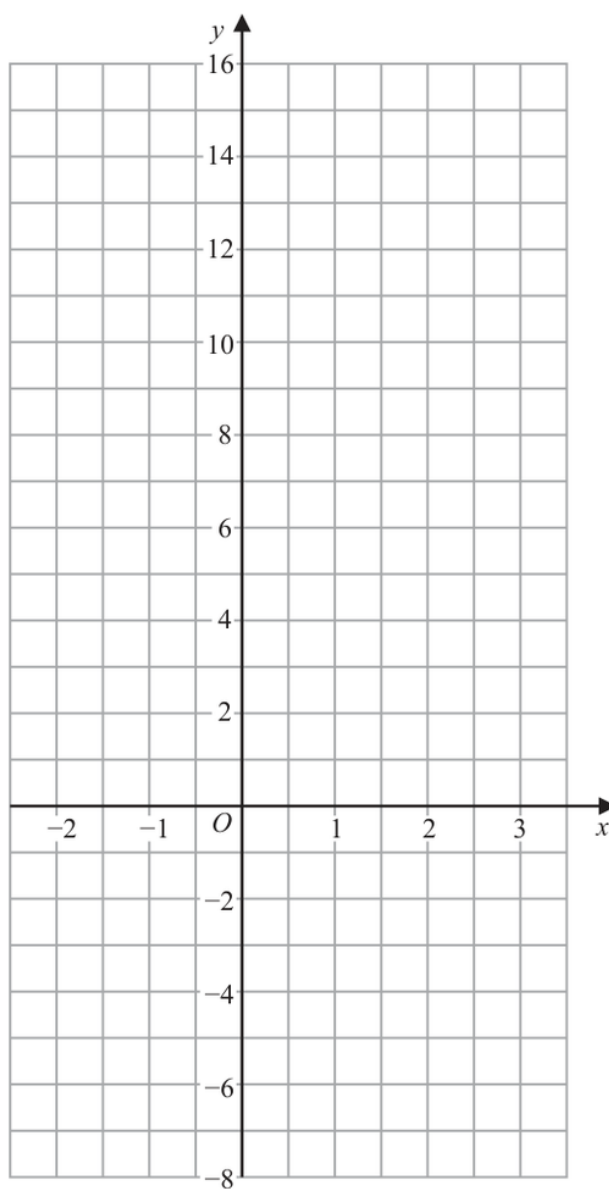
FINAL ANSWER

45.

PAST PAPER QUESTION**Q#: 3****Marks: 3****TOPIC: Linear Graphs ($y = mx + c$)**

May-Nov 2020 P1H - MayNov 2020 | P1H

- 3 On the grid, draw the graph of $y = 7 - 4x$ for values of x from -2 to 3



(Total for Question 3 is 3 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 3****TOPIC: Linear Graphs ($y = mx + c$)**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Read the graph**

$$y = 7 - 4x$$

2. Useful points are

Useful points are:

$$(-2, 15), \quad (0, 7), \quad (1, 3), \quad (2, -1), \quad (3, -5)$$

3. Plot these points and join

Plot these points and join them with a straight line.

FINAL ANSWERThe graph is the straight line $y = 7 - 4x$.

PAST PAPER QUESTION**Q#: 4****Marks: 4****TOPIC: Statistics Toolkit**

May-Nov 2020 P1H - MayNov 2020 | P1H

- 4 Here is a list of six numbers written in order of size.

4 7 x 10 y y

The numbers have

a median of 9

a mean of 11

Find the value of x and the value of y .

$x =$

$y =$

(Total for Question 4 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 4****Marks: 4****TOPIC: Statistics Toolkit**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The median is**

The median is 9:

$$\frac{x + 10}{2} = 9$$

$$x = 8$$

2. Calculate statistic

The mean is 11, so the total is

$$6 \times 11 = 66$$

$$4 + 7 + 8 + 10 + y + y = 66$$

$$29 + 2y = 66$$

$$2y = 37$$

$$y = 18.5$$

FINAL ANSWER

$$x = 8, y = 18.5.$$

PAST PAPER QUESTION**Q#: 5****Marks: 4****TOPIC: Powers, Roots & Standard Form**

May-Nov 2020 P1H - MayNov 2020 | P1H

5 (a) Write 5.7×10^{-3} as an ordinary number.

.....
(1)

(b) Write 800 000 in standard form.

.....
(1)

(c) Work out $\frac{3 \times 10^5 - 2.7 \times 10^4}{6 \times 10^{-2}}$

.....
(2)

(Total for Question 5 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 5****Marks: 4****TOPIC: Powers, Roots & Standard Form**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Convert standard form**

$$5.7 \times 10^{-3} = 0.0057$$

$$800000 = 8 \times 10^5$$

2. Convert standard form

For part (c), first simplify the numerator:

$$3 \times 10^5 - 2.7 \times 10^4 = 300000 - 27000 = 273000$$

3. Convert standard form

Then

$$\frac{273000}{6 \times 10^{-2}} = \frac{273000}{0.06} = 4550000$$

$$4550000 = 4.55 \times 10^6$$

FINAL ANSWER(a) 0.0057, (b) 8×10^5 , (c) 4.55×10^6

PAST PAPER QUESTION**Q#: 6****Marks: 3****TOPIC: Standard & Compound Units**

May-Nov 2020 P1H - MayNov 2020 | P1H

- 6** A rocket travelled 100 km at an average speed of 28 440 km/h.

Work out how long it took the rocket to travel the 100 km.

Give your answer in seconds, correct to the nearest second.

..... seconds

(Total for Question 6 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 6****Marks: 3****TOPIC: Standard & Compound Units**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use**

Use

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

$$\text{time} = \frac{100}{28440} \text{ hours}$$

2. Convert hours to seconds

Convert hours to seconds:

$$\frac{100}{28440} \times 3600 = 12.658 \dots$$

3. To the nearest second

To the nearest second,

$$12.658 \dots \approx 13$$

FINAL ANSWER

13 seconds.

PAST PAPER QUESTION**Q#: 7****Marks: 8****TOPIC: Factorising**

May-Nov 2020 P1H - MayNov 2020 | P1H

- 7 (a) Solve $5(4 - x) = 7 - 3x$
Show clear algebraic working.

$$x = \dots\dots\dots$$

(3)

- (b) Factorise fully $16m^3g^3 + 24m^2g^5$

$$\dots\dots\dots$$

(2)

- (c) (i) Factorise $y^2 - 2y - 48$

$$\dots\dots\dots$$

(2)

- (ii) Hence, solve $y^2 - 2y - 48 = 0$

$$\dots\dots\dots$$

(1)

(Total for Question 7 is 8 marks)

PAST PAPER SOLUTION**Q#: 7****Marks: 8****TOPIC: Factorising**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$5(4 - x) = 7 - 3x$$

$$20 - 5x = 7 - 3x$$

$$13 = 2x$$

$$x = \frac{13}{2}$$

2. (b)

(b)

$$16m^3g^3 + 24m^2g^5 = 8m^2g^3(2m + 3g^2)$$

3. (c)(i)

(c)(i)

$$y^2 - 2y - 48 = (y - 8)(y + 6)$$

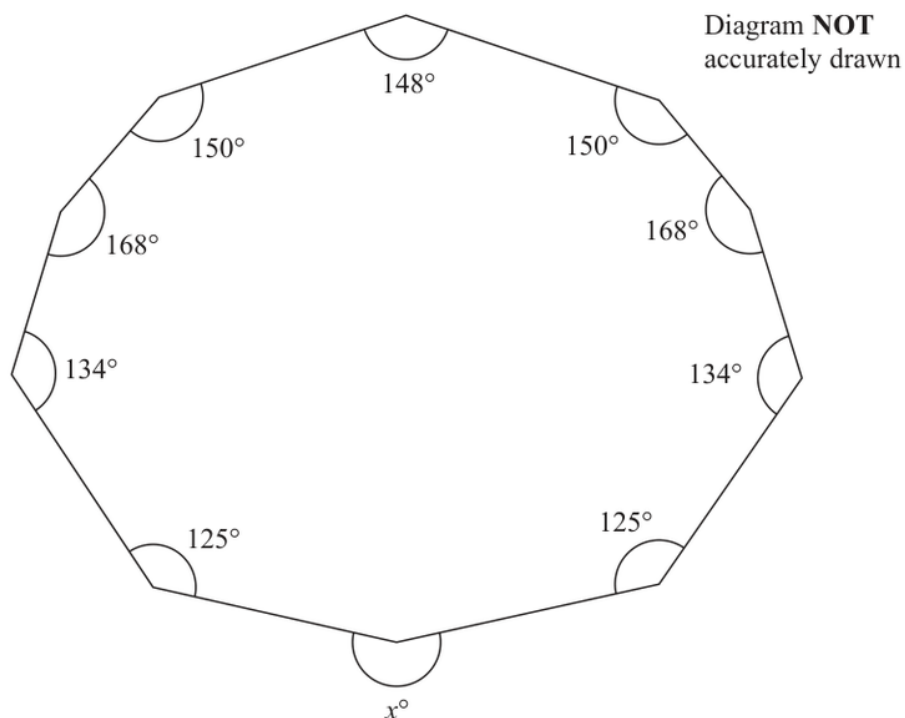
4. Factorise expression(c)(ii) $y = 8$ or $y = -6$.**FINAL ANSWER**

$$x = \frac{13}{2}, 8m^2g^3(2m + 3g^2), (y - 8)(y + 6), y = 8 \text{ or } y = -6$$

PAST PAPER QUESTION**Q#: 8****Marks: 4****TOPIC: Angles in Polygons & Parallel Lines**

May-Nov 2020 P1H - MayNov 2020 | P1H

8 Here is a 10-sided polygon.



Work out the value of x .

 $x = \dots\dots\dots$

(Total for Question 8 is 4 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 4****TOPIC: Angles in Polygons & Parallel Lines**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The sum of the interior**

The sum of the interior angles of a 10-sided polygon is

$$(10 - 2) \times 180 = 1440^\circ$$

2. The sum of the 9

The sum of the 9 given interior angles is

$$150 + 168 + 134 + 125 + 125 + 134 + 168 + 150 + 148 = 1302^\circ$$

3. Find the gradient

So the missing interior angle is

$$1440^\circ - 1302^\circ = 138^\circ$$

4. The angle is the reflexThe angle x is the reflex angle outside the polygon at this vertex:

$$x = 360^\circ - 138^\circ = 222^\circ$$

FINAL ANSWER

222°.

PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Percentages**

May-Nov 2020 P1H - MayNov 2020 | P1H

9 In a sale, normal prices are reduced by 20%

A bag costs 1080 rupees in the sale.

Work out the normal price of the bag.

..... rupees

(Total for Question 9 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 9****Marks: 3****TOPIC: Percentages**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. A reduction of means the**

A reduction of 20% means the sale price is 80% of the normal price.

$$0.80 \times \text{normal price} = 1080$$

$$\text{normal price} = \frac{1080}{0.80} = 1350$$

FINAL ANSWER

1350 rupees

EA

PAST PAPER QUESTION**Q#: 10****Marks: 3****TOPIC: Prime Factors, HCF & LCM**

May-Nov 2020 P1H - MayNov 2020 | P1H

10 $A = 2 \times 3^{43}$
 $B = 16 \times 3^{37}$

- (a) Find the highest common factor (HCF) of A and B .

.....
(1)

- (b) Express the number $A \times B$ as a product of powers of its prime factors.
Give your answer in its simplest form.

.....
(2)

.....
(Total for Question 10 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 10****Marks: 3****TOPIC: Prime Factors, HCF & LCM**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the LCM**

$$A = 2 \times 3^{43}$$

$$B = 16 \times 3^{37} = 2^4 \times 3^{37}$$

2. Find the LCM

For the HCF, take the smaller powers:

$$\text{HCF} = 2 \times 3^{37}$$

3. Find the LCMFor $A \times B$,

$$A \times B = (2 \times 3^{43})(2^4 \times 3^{37})$$

$$A \times B = 2^5 \times 3^{80}$$

FINAL ANSWER(a) 2×3^{37} , (b) $2^5 \times 3^{80}$

PAST PAPER QUESTION**Q#: 11****Marks: 6****TOPIC: Area & Perimeter**

May-Nov 2020 P1H - MayNov 2020 | P1H

- 11** The diagram shows trapezium $ABCD$ in which BC and AD are parallel.

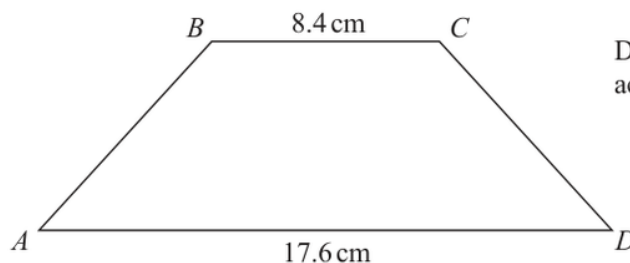


Diagram **NOT**
accurately drawn

The trapezium has exactly one line of symmetry.

$$BC = 8.4 \text{ cm}$$

$$AD = 17.6 \text{ cm}$$

The trapezium has area 179.4 cm^2

Work out the size of angle ABC .

Give your answer correct to 1 decimal place.

o

(Total for Question 11 is 6 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 6****TOPIC: Area & Perimeter**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The average of the parallel**

The average of the parallel sides is

$$\frac{8.4 + 17.6}{2} = 13$$

2. Find the gradient

So the perpendicular height is

$$179.4 \div 13 = 13.8$$

3. The horizontal offset on each

The horizontal offset on each side is

$$\frac{17.6 - 8.4}{2} = 4.6$$

4. Use trigonometryThe acute angle between AB and the horizontal is

$$\tan^{-1} \left(\frac{13.8}{4.6} \right) = 71.565 \dots^\circ$$

5. Use trigonometrySo the interior angle ABC is

$$180^\circ - 71.565 \dots^\circ = 108.434 \dots^\circ$$

FINAL ANSWER

108.4°.

PAST PAPER QUESTION**Q#: 12****Marks: 4****TOPIC: Simultaneous Equations**

May-Nov 2020 P1H - MayNov 2020 | P1H

12 Solve the simultaneous equations

$$7x - 2y = 34$$

$$3x + 5y = -3$$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

(Total for Question 12 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 12****Marks: 4****TOPIC: Simultaneous Equations**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve simultaneous equations**

$$7x - 2y = 34$$

$$3x + 5y = -3$$

2. Multiply the first equation by

Multiply the first equation by 5:

$$35x - 10y = 170$$

3. Multiply the second equation by

Multiply the second equation by 2:

$$6x + 10y = -6$$

4. Add

Add:

$$41x = 164$$

$$x = 4$$

5. Solve simultaneous equationsSubstitute into $3x + 5y = -3$:

$$12 + 5y = -3$$

$$y = -3$$

FINAL ANSWER

$$x = 4, y = -3.$$

PAST PAPER QUESTION**Q#: 13****Marks: 3****TOPIC: Compound Interest & Depreciation**

May-Nov 2020 P1H - MayNov 2020 | P1H

- 13** Jan invests \$8000 in a savings account.
The account pays compound interest at a rate of $x\%$ per year.
At the end of 6 years, there is a total of \$8877.62 in the account.
Work out the value of x .
Give your answer correct to 2 decimal places.

 $x = \dots\dots\dots$

(Total for Question 13 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 13****Marks: 3****TOPIC: Compound Interest & Depreciation**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the compound interest rate**Let the compound interest rate be $x\%$.

$$8000 \left(1 + \frac{x}{100}\right)^6 = 8877.62$$

$$1 + \frac{x}{100} = \sqrt[6]{\frac{8877.62}{8000}} = 1.017500\dots$$

$$x = 1.7500\dots$$

2. Use compound interest

Correct to 2 decimal places,

$$x = 1.75$$

FINAL ANSWER

$$x = 1.75$$

PAST PAPER QUESTION**Q#: 14****Marks: 3****TOPIC: Direct & Inverse Proportion**

May-Nov 2020 P1H - MayNov 2020 | P1H

14 F is inversely proportional to the square of v .

Given that $F = 6.5$ when $v = 4$

find a formula for F in terms of v .

(Total for Question 14 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 14****Marks: 3****TOPIC: Direct & Inverse Proportion**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find inverse function** F is inversely proportional to v^2 , so

$$F = \frac{k}{v^2}$$

2. Use whenUse $F = 6.5$ when $v = 4$:

$$6.5 = \frac{k}{4^2}$$

$$k = 6.5 \times 16 = 104$$

3. Find inverse function

Therefore

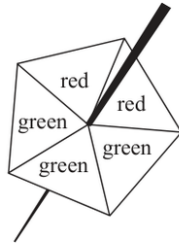
$$F = \frac{104}{v^2}$$

FINAL ANSWER

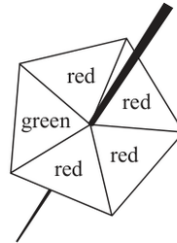
$$F = \frac{104}{v^2}.$$

PAST PAPER QUESTION**Q#: 15****Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

May-Nov 2020 P1H - MayNov 2020 | P1H

15 Harry has two fair 5-sided spinners.

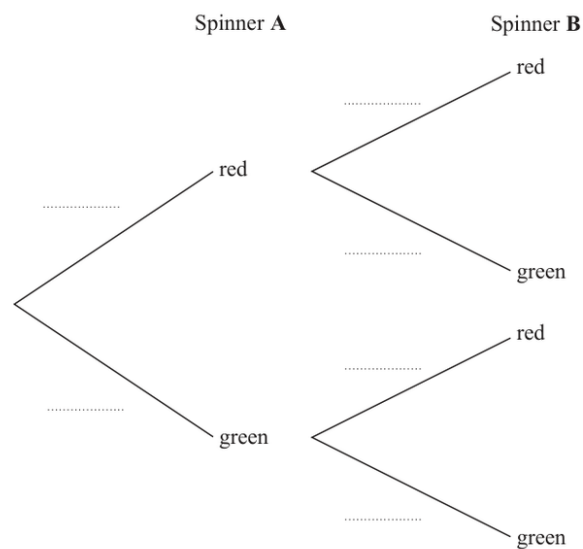
Spinner A



Spinner B

Harry is going to spin each spinner once.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that at least one of the spinners will land on green.

(3)

(Total for Question 15 is 5 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Spinner A has red sectors**

Spinner A has 2 red sectors and 3 green sectors:

$$P(A_R) = \frac{2}{5}, \quad P(A_G) = \frac{3}{5}$$

2. Spinner B has red sectors

Spinner B has 4 red sectors and 1 green sector:

$$P(B_R) = \frac{4}{5}, \quad P(B_G) = \frac{1}{5}$$

3. At least one green is

At least one green is the complement of both red.

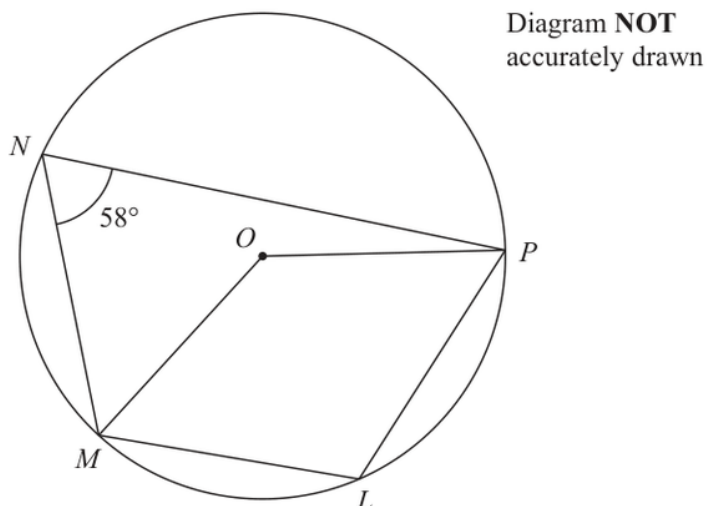
$$\begin{aligned} P(\text{at least one green}) &= 1 - \frac{2}{5} \times \frac{4}{5} \\ &= 1 - \frac{8}{25} = \frac{17}{25} \end{aligned}$$

FINAL ANSWER

$$\frac{17}{25}.$$

PAST PAPER QUESTION**Q#: 16** **Marks: 4****TOPIC: Circle Theorems**

May-Nov 2020 P1H - MayNov 2020 | P1H

16

L, M, N and P are points on a circle, centre O

Angle $MNP = 58^\circ$

(a) (i) Find the size of angle MLP

°

(ii) Give a reason for your answer.

(2)

(b) Find the size of the reflex angle MOP

°

(2)

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 4****TOPIC: Circle Theorems**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a) Opposite angles in cyclic**(a) Opposite angles in cyclic quadrilateral $MNPL$ add to 180° :

$$\angle MLP = 180^\circ - 58^\circ = 122^\circ$$

2. (b) The angle at the(b) The angle at the centre is twice the angle at the circumference standing on the same arc MP :

$$\angle MOP = 2 \times 58^\circ = 116^\circ$$

3. The reflex angle is

The reflex angle is

$$360^\circ - 116^\circ = 244^\circ$$

FINAL ANSWER $\angle MLP = 122^\circ$, and reflex $\angle MOP = 244^\circ$.

PAST PAPER QUESTION**Q#: 17****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

May-Nov 2020 P1H - MayNov 2020 | P1H

- 17** A metal block has a mass of 5 kg, correct to the nearest 50 grams.
The block has a volume of $(1.84 \times 10^{-3})\text{m}^3$, correct to 3 significant figures.

Work out the upper bound for the density of the block.
Give your answer in kg/m^3 correct to 1 decimal place.
Show your working clearly.

..... kg/m^3

(Total for Question 17 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 17****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Density is**

Density is

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

2. To get the upper bound

To get the upper bound for density, use the upper bound for mass and the lower bound for volume.

3. The mass is kg correct

The mass is 5 kg correct to the nearest 50 g.

$$50 \text{ g} = 0.05 \text{ kg}$$

4. Find upper bound

Half of this is 0.025 kg, so

$$\text{upper bound for mass} = 5 + 0.025 = 5.025$$

5. Find upper boundThe volume is $1.84 \times 10^{-3} = 0.00184 \text{ m}^3$, correct to 3 significant figures.**6. The lower bound is**

The lower bound is

$$0.00184 - 0.000005 = 0.001835$$

7. Find upper bound

So the upper bound for the density is

$$\frac{5.025}{0.001835} = 2738.419 \dots$$

8. Find upper bound

Correct to 1 decimal place,

$$2738.4$$

FINAL ANSWER

$$2738.4 \text{ kg/m}^3$$

PAST PAPER QUESTION

Q#: 18 Marks: 5

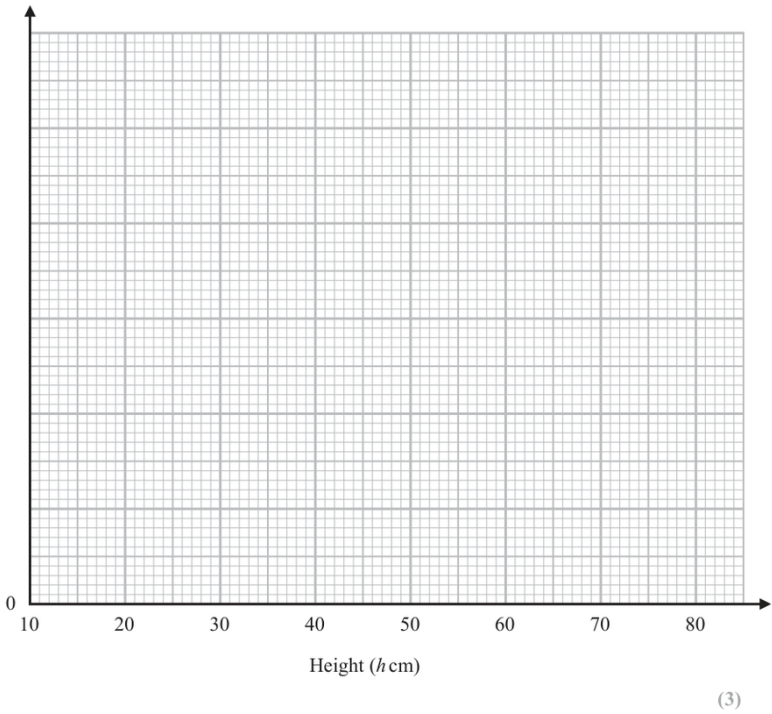
TOPIC: Histograms

May-Nov 2020 P1H - MayNov 2020 | P1H

18 The table gives information about the heights, in centimetres, of some plants.

Height (h cm)	Frequency
$10 < h \leq 20$	35
$20 < h \leq 35$	45
$35 < h \leq 50$	75
$50 < h \leq 70$	40
$70 < h \leq 80$	8

(a) On the grid, draw a histogram for this information.



(b) Work out an estimate for the number of these plants with a height greater than 40 cm.

.....

(Total for Question 18 is 5 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 5****TOPIC: Histograms**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The histogram densities are**

The histogram densities are

$$\frac{35}{10} = 3.5, \quad \frac{45}{15} = 3, \quad \frac{75}{15} = 5, \quad \frac{40}{20} = 2, \quad \frac{8}{10} = 0.8$$

2. Use histogram

For plants taller than 40 cm:

3. Use histogramFrom $40 < h \leq 50$:

$$10(5) = 50$$

4. Use histogramFrom $50 < h \leq 70$:

$$40$$

5. Use histogramFrom $70 < h \leq 80$:

$$8$$

$$50 + 40 + 8 = 98$$

FINAL ANSWER

draw bars with densities 3.5, 3, 5, 2, 0.8, and 98 plants.

PAST PAPER QUESTION**Q#: 19****Marks: 3****TOPIC: Surds**

May-Nov 2020 P1H - MayNov 2020 | P1H

19 Without using a calculator, rationalise the denominator of $\frac{6}{3 - \sqrt{7}}$

Simplify your answer.

You must show each stage of your working.

(Total for Question 19 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 19****Marks: 3****TOPIC: Surds**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify surd**

$$\frac{6}{3 - \sqrt{7}}$$

2. Rationalise the denominator

Rationalise the denominator:

$$\begin{aligned}\frac{6}{3 - \sqrt{7}} \times \frac{3 + \sqrt{7}}{3 + \sqrt{7}} &= \frac{6(3 + \sqrt{7})}{9 - 7} \\ &= \frac{6(3 + \sqrt{7})}{2} = 3(3 + \sqrt{7}) \\ &= 9 + 3\sqrt{7}\end{aligned}$$

FINAL ANSWER

$$9 + 3\sqrt{7}$$

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

May-Nov 2020 P1H - MayNov 2020 | P1H

20 **R** and **S** are two similar solid shapes.

Shape **R** has surface area 108 cm^2 and volume 135 cm^3

Shape **S** has surface area 300 cm^2

Work out the volume of shape **S**.

..... cm^3

(Total for Question 20 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Shape R has surface area**Shape R has surface area 108 cm^2 and volume 135 cm^3 .**2. Shape S has surface area**Shape S has surface area 300 cm^2 .**3. Surface area ratio S R**

Surface area ratio S:R:

$$300 : 108 = 25 : 9$$

4. Split the ratio

So the linear ratio S:R is

$$5 : 3$$

5. Split the ratio

Therefore the volume ratio S:R is

$$5^3 : 3^3 = 125 : 27$$

6. Volume of S

Volume of S:

$$135 \times \frac{125}{27} = 625$$

FINAL ANSWER**625 cm³**

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

May-Nov 2020 P1H - MayNov 2020 | P1H

20 **R** and **S** are two similar solid shapes.

Shape **R** has surface area 108 cm^2 and volume 135 cm^3

Shape **S** has surface area 300 cm^2

Work out the volume of shape **S**.

..... cm^3

(Total for Question 20 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Area & Volume of Similar Shapes**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Shape R has surface area**Shape R has surface area 108 cm^2 and volume 135 cm^3 .**2. Shape S has surface area**Shape S has surface area 300 cm^2 .**3. Surface area ratio S R**

Surface area ratio S:R:

$$300 : 108 = 25 : 9$$

4. Split the ratio

So the linear ratio S:R is

$$5 : 3$$

5. Split the ratio

Therefore the volume ratio S:R is

$$5^3 : 3^3 = 125 : 27$$

6. Volume of S

Volume of S:

$$135 \times \frac{125}{27} = 625$$

FINAL ANSWER**625 cm³**

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Algebraic Fractions**

May-Nov 2020 P1H - MayNov 2020 | P1H

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: Algebraic Fractions**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

2. Factor

Factor:

$$9x^2 - 4 = (3x - 2)(3x + 2)$$

$$3x^2 - 13x - 10 = (3x + 2)(x - 5)$$

3. Use trigonometry

So the first product becomes

$$\frac{1}{3x-2} \times \frac{(3x-2)(3x+2)}{(3x+2)(x-5)}$$

4. Cancel common factors

Cancel common factors:

$$\frac{1}{x-5}$$

5. Use trigonometry

Now subtract:

$$\frac{1}{x-5} - \frac{7}{x-1}$$

6. Use the common denominatorUse the common denominator $(x-5)(x-1)$:

$$\begin{aligned} & \frac{x-1-7(x-5)}{(x-5)(x-1)} \\ &= \frac{x-1-7x+35}{(x-5)(x-1)} \\ &= \frac{34-6x}{(x-5)(x-1)} \end{aligned}$$

FINAL ANSWER

$$\frac{34 - 6x}{(x - 5)(x - 1)}$$

EA

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Algebraic Fractions**

May-Nov 2020 P1H - MayNov 2020 | P1H

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

(Total for Question 21 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: Algebraic Fractions**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

2. Factor

Factor:

$$9x^2 - 4 = (3x - 2)(3x + 2)$$

$$3x^2 - 13x - 10 = (3x + 2)(x - 5)$$

3. Use trigonometry

So the first product becomes

$$\frac{1}{3x-2} \times \frac{(3x-2)(3x+2)}{(3x+2)(x-5)}$$

4. Cancel common factors

Cancel common factors:

$$\frac{1}{x-5}$$

5. Use trigonometry

Now subtract:

$$\frac{1}{x-5} - \frac{7}{x-1}$$

6. Use the common denominatorUse the common denominator $(x-5)(x-1)$:

$$\begin{aligned} & \frac{x-1-7(x-5)}{(x-5)(x-1)} \\ &= \frac{x-1-7x+35}{(x-5)(x-1)} \\ &= \frac{34-6x}{(x-5)(x-1)} \end{aligned}$$

FINAL ANSWER

$$\frac{34 - 6x}{(x - 5)(x - 1)}$$

EA

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Coordinate Geometry**

May-Nov 2020 P1H - MayNov 2020 | P1H

22 $ABCD$ is a rhombus.The diagonals, AC and BD , intersect at the point M .The coordinates of M are $(6, -11)$ The points A and C both lie on the line with equation $2y + 7x = 20$ Find the exact coordinates of the point where the line through B and D intersects the y -axis.

(.....,)

(Total for Question 22 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 22****Marks: 4****TOPIC: Coordinate Geometry**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the gradient**

In a rhombus, the diagonals are perpendicular and bisect each other.

2. The diagonal lies on

The diagonal AC lies on

$$2y + 7x = 20$$

$$2y = -7x + 20$$

$$y = -\frac{7}{2}x + 10$$

3. Find the gradient

So the gradient of AC is

$$-\frac{7}{2}$$

4. Find the gradient

Therefore the gradient of BD is

$$\frac{2}{7}$$

5. Calculate value

The line BD passes through

$$M = (6, -11)$$

6. Evaluate fraction

So

$$y + 11 = \frac{2}{7}(x - 6)$$

7. To find where this line

To find where this line intersects the y -axis, put $x = 0$:

$$y + 11 = \frac{2}{7}(0 - 6)$$

$$y + 11 = -\frac{12}{7}$$

$$y = -11 - \frac{12}{7}$$

$$y = -\frac{89}{7}$$

FINAL ANSWER

$$\left(0, -\frac{89}{7}\right)$$

EA

PAST PAPER QUESTION**Q#: 22****Marks: 4****TOPIC: Coordinate Geometry**

May-Nov 2020 P1H - MayNov 2020 | P1H

22 $ABCD$ is a rhombus.The diagonals, AC and BD , intersect at the point M .The coordinates of M are $(6, -11)$ The points A and C both lie on the line with equation $2y + 7x = 20$ Find the exact coordinates of the point where the line through B and D intersects the y -axis.

(..... ,)

(Total for Question 22 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 22****Marks: 4****TOPIC: Coordinate Geometry**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find the gradient**

In a rhombus, the diagonals are perpendicular and bisect each other.

2. The diagonal lies on

The diagonal AC lies on

$$2y + 7x = 20$$

$$2y = -7x + 20$$

$$y = -\frac{7}{2}x + 10$$

3. Find the gradient

So the gradient of AC is

$$-\frac{7}{2}$$

4. Find the gradient

Therefore the gradient of BD is

$$\frac{2}{7}$$

5. Calculate value

The line BD passes through

$$M = (6, -11)$$

6. Evaluate fraction

So

$$y + 11 = \frac{2}{7}(x - 6)$$

7. To find where this line

To find where this line intersects the y -axis, put $x = 0$:

$$y + 11 = \frac{2}{7}(0 - 6)$$

$$y + 11 = -\frac{12}{7}$$

$$y = -11 - \frac{12}{7}$$

$$y = -\frac{89}{7}$$

FINAL ANSWER

$$\left(0, -\frac{89}{7}\right)$$

EA

PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Differentiation**

May-Nov 2020 P1H - MayNov 2020 | P1H

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x , in terms of p and m , for which the gradient of **C** is negative.

(Total for Question 23 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 4****TOPIC: Differentiation**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Differentiate**

$$y = px^3 - mx$$

2. Differentiate

Differentiate:

$$\frac{dy}{dx} = 3px^2 - m$$

3. The gradient is negative when

The gradient is negative when

$$3px^2 - m < 0$$

$$3px^2 < m$$

$$x^2 < \frac{m}{3p}$$

4. Simplify surd

So

$$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$$

FINAL ANSWER

$$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$$

PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Differentiation**

May-Nov 2020 P1H - MayNov 2020 | P1H

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x , in terms of p and m , for which the gradient of **C** is negative.

(Total for Question 23 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 4****TOPIC: Differentiation**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Differentiate**

$$y = px^3 - mx$$

2. Differentiate

Differentiate:

$$\frac{dy}{dx} = 3px^2 - m$$

3. The gradient is negative when

The gradient is negative when

$$3px^2 - m < 0$$

$$3px^2 < m$$

$$x^2 < \frac{m}{3p}$$

4. Simplify surd

So

$$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$$

FINAL ANSWER

$$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$$

PAST PAPER QUESTION**Q#: 24****Marks: 4****TOPIC: Sequences**

May-Nov 2020 P1H - MayNov 2020 | P1H

24 Here are the first five terms of an arithmetic sequence.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

(Total for Question 24 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 24****Marks: 4****TOPIC: Sequences**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The arithmetic sequence is**

The arithmetic sequence is

$$8, 15, 22, 29, 36, \dots$$

2. Calculate value

So

$$a = 8, \quad d = 7$$

3. The n th term isThe n th term is

$$u_n = 8 + 7(n - 1) = 7n + 1$$

$$u_{50} = 7(50) + 1 = 351$$

$$u_{100} = 7(100) + 1 = 701$$

4. There are

There are

$$100 - 50 + 1 = 51$$

5. Find the n th term

terms from the 50th to the 100th inclusive.

$$\text{sum} = \frac{51}{2}(351 + 701)$$

$$= 26826$$

FINAL ANSWER

26826

PAST PAPER QUESTION**Q#: 24****Marks: 4****TOPIC: Sequences**

May-Nov 2020 P1H - MayNov 2020 | P1H

24 Here are the first five terms of an arithmetic sequence.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

(Total for Question 24 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 24****Marks: 4****TOPIC: Sequences**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The arithmetic sequence is**

The arithmetic sequence is

$$8, 15, 22, 29, 36, \dots$$

2. Calculate value

So

$$a = 8, \quad d = 7$$

3. The n th term isThe n th term is

$$u_n = 8 + 7(n - 1) = 7n + 1$$

$$u_{50} = 7(50) + 1 = 351$$

$$u_{100} = 7(100) + 1 = 701$$

4. There are

There are

$$100 - 50 + 1 = 51$$

5. Find the n th term

terms from the 50th to the 100th inclusive.

$$\text{sum} = \frac{51}{2}(351 + 701)$$

$$= 26826$$

FINAL ANSWER

26826

PAST PAPER QUESTION**Q#: 25****Marks: 3****TOPIC: Transformations of Graphs**

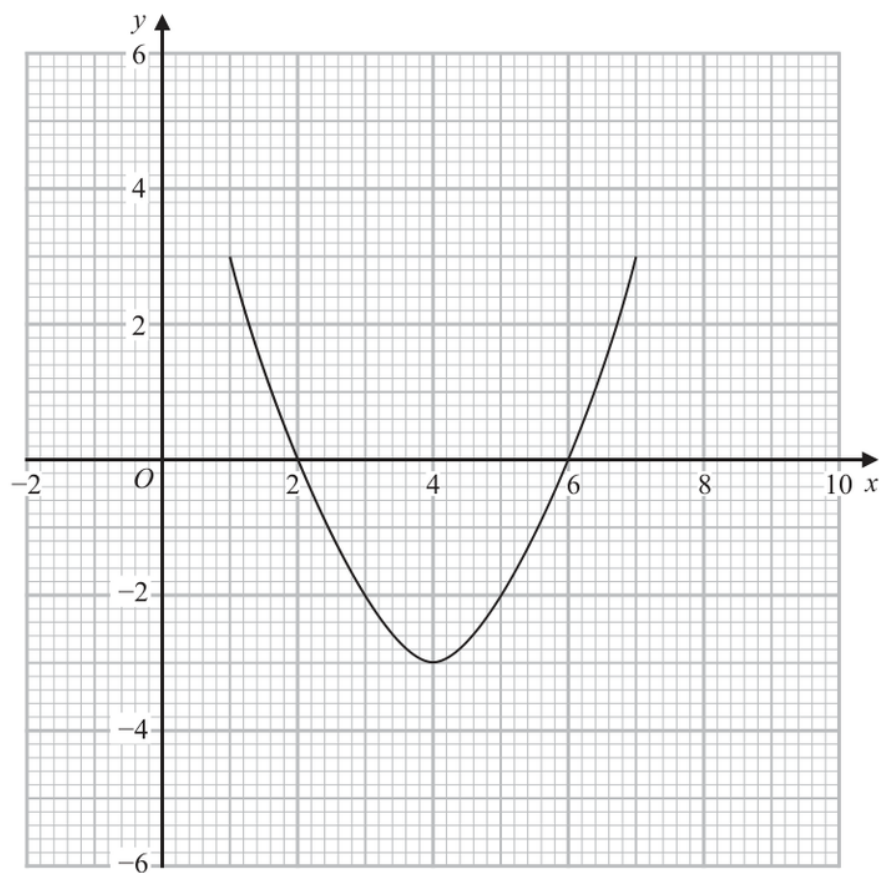
May-Nov 2020 P1H - MayNov 2020 | P1H

- 25** The curve with equation $y = g(x)$ is transformed to the curve with equation $y = -g(x)$ by the single transformation **T**.

(a) Describe fully the transformation **T**.

(1)

The diagram shows the graph of $y = f(x)$



(b) On the grid, draw the graph of $y = 2f(x - 1)$

(2)

(Total for Question 25 is 3 marks)

PAST PAPER SOLUTION**Q#: 25****Marks: 3****TOPIC: Transformations of Graphs**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Apply transformation**

For part (a),

$$y = g(x) \rightarrow y = -g(x)$$

2. The coordinates stay the sameThe x -coordinates stay the same and the y -coordinates change sign.**3. Apply transformation**So the transformation is a reflection in the x -axis.**4. Calculate value**

For part (b),

$$y = 2f(x - 1)$$

5. Read the graphFirst, $x - 1$ moves the graph 1 unit right. Then the factor 2 stretches the graph vertically by scale factor 2.**6. Use points from the original**

Use points from the original graph:

$$(1, 3) \rightarrow (2, 6)$$

$$(2, 0) \rightarrow (3, 0)$$

$$(4, -3) \rightarrow (5, -6)$$

$$(6, 0) \rightarrow (7, 0)$$

$$(7, 3) \rightarrow (8, 6)$$

7. Apply transformation

Draw the transformed parabola through these points.

FINAL ANSWERreflection in the x -axis; transformed graph through $(2, 6)$, $(3, 0)$, $(5, -6)$, $(7, 0)$, $(8, 6)$

PAST PAPER QUESTION**Q#: 25** **Marks: 3****TOPIC: Transformations of Graphs**

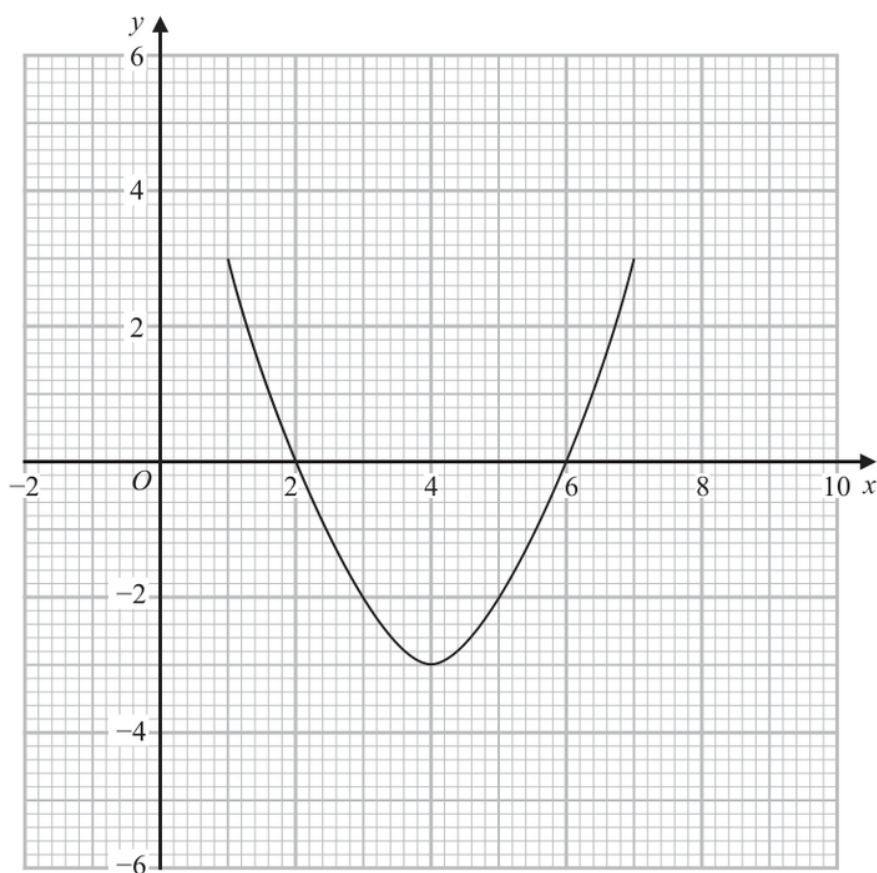
May-Nov 2020 P1H - MayNov 2020 | P1H

- 25** The curve with equation $y = g(x)$ is transformed to the curve with equation $y = -g(x)$ by the single transformation **T**.

(a) Describe fully the transformation **T**.

(1)

The diagram shows the graph of $y = f(x)$



(b) On the grid, draw the graph of $y = 2f(x - 1)$

(2)

(Total for Question 25 is 3 marks)

PAST PAPER SOLUTION**Q#: 25****Marks: 3****TOPIC: Transformations of Graphs**

May-Nov 2020 P1H - MayNov 2020 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Apply transformation**

For part (a),

$$y = g(x) \rightarrow y = -g(x)$$

2. The coordinates stay the sameThe x -coordinates stay the same and the y -coordinates change sign.**3. Apply transformation**So the transformation is a reflection in the x -axis.**4. Calculate value**

For part (b),

$$y = 2f(x - 1)$$

5. Read the graphFirst, $x - 1$ moves the graph 1 unit right. Then the factor 2 stretches the graph vertically by scale factor 2.**6. Use points from the original**

Use points from the original graph:

$$(1, 3) \rightarrow (2, 6)$$

$$(2, 0) \rightarrow (3, 0)$$

$$(4, -3) \rightarrow (5, -6)$$

$$(6, 0) \rightarrow (7, 0)$$

$$(7, 3) \rightarrow (8, 6)$$

7. Apply transformation

Draw the transformed parabola through these points.

FINAL ANSWERreflection in the x -axis; transformed graph through $(2, 6)$, $(3, 0)$, $(5, -6)$, $(7, 0)$, $(8, 6)$